TEXORA · LECTURE SHEET

Recipe Prediction for Color Matching

Colour matching is prediction with three unknowns at once. Kubelka–Munk additivity turns the reflectance spectrum of a standard into a solvable system — this sheet works the mathematics by hand, once, so the computer's answer is never a mystery.

WP-3207 · ADVANCED MACHINERIES AND RELEVANT CALCULATIONS · TOPIC 3 · SPECIALIZATION — WET PROCESSING

1. From reflectance to arithmetic: the Kubelka–Munk function

A spectrophotometer reports reflectance R(λ) — the fraction of incident light returned at each wavelength. Reflectance itself does not add up over dyes, but the Kubelka–Munk ratio does:

$$K/S = (1 - R)^2 / 2R$$

K (absorption) grows in direct proportion to the amount of colourant present, while S (scattering) is set almost entirely by the substrate. On a fixed substrate, therefore, $\mathbf{(K/S)_{mixture} \approx \Sigma\, c_i \times (K/S)_{unit,\ i}}$ at every wavelength, where c_i is the concentration of dye i and $(K/S)_{unit,\ i}$ is that dye's calibrated contribution per 1% owf (Topic 1's build-up series, measured at each wavelength). This additivity is the entire basis of computer colour matching.

2. Setting up the match: three dyes, three wavelengths

With three dyes there are three unknown concentrations, so three independent equations suffice in principle — taken at three wavelengths where the dyes' contributions differ most. (Production systems use all 16–31 measured wavelengths and least-squares fitting, which averages out measurement noise and additivity error; the 3 × 3 system below is the same mathematics with the noise removed for teaching.)

Λ (NM)	YELLOW, K/S PER 1%	RED, K/S PER 1%	BLUE, K/S PER 1%	STANDARD K/S
420	2.60	0.35	0.20	3.10
520	0.55	2.30	0.45	2.60
620	0.25	0.60	2.10	1.70

The standard's K/S values come from its measured reflectance: at 420 nm, for instance, K/S = 3.10 corresponds to R ≈ 0.122 (solving $(1-R)^2 = 2R \times 3.10$). The equations are:

$$2.60Y + 0.35R + 0.20B = 3.10 \quad \cdot \quad 0.55Y + 2.30R + 0.45B = 2.60 \quad \cdot \quad 0.25Y + 0.60R + 2.10B = 1.70$$

WORKED EXAMPLE 1 — SOLVING THE TRICHROMATIC SYSTEM

Solving the three equations simultaneously (elimination, or the match-prediction software) gives: **Yellow = 1.05% owf, Red = 0.79% owf, Blue = 0.46% owf** (total 2.30% owf).
Verification at 520 nm: 0.55 × 1.05 + 2.30 × 0.79 + 0.45 × 0.46 = 0.578 + 1.817 + 0.207 = 2.60 ✓ — and the same check closes at 420 and 620 nm. Always re-substitute: a predicted recipe that does not reproduce its own equations was solved wrongly, not approximately.

3. Judging the match — ΔE and the correction loop

The trial dyeing is measured and compared with the standard in CIELAB space. The colour difference is:

$$\Delta E^*_{ab} = \sqrt{(\Delta L^{*2} + \Delta a^{*2} + \Delta b^{*2})}$$

WORKED EXAMPLE 2 — IS THIS TRIAL ACCEPTABLE?

Standard: L* 45.2, a* 28.4, b* −12.6. Trial: L* 46.0, a* 27.9, b* −13.1.
ΔL* = −0.8 (trial lighter), Δa* = +0.5, Δb* = +0.5.
$\Delta E = \sqrt{(0.64 + 0.25 + 0.25)} = \sqrt{1.14} =$ **1.07**.
Against a tolerance of ΔE ≤ 1.0 the trial *just* fails: it is lighter and slightly less red. Correction: deepen all three dyes slightly (lightness error dominates) with a small relative increase in the red — then re-dye rather than "correcting on paper", because additivity error grows with every paper correction.

Correction rules of thumb: a lightness error (ΔL) is corrected by scaling the whole recipe; a hue error (Δa, Δb) by trading concentration between the two dyes whose hues bracket the error; a chroma error by scaling the dye that owns the dominant wavelength region. Modern systems compute the correction from the same unit K/S data by least squares — a **metameric** set of solutions exists whenever more dyes are offered than needed, and the system should choose among them by cost and metamerism, not by fit alone.

4. Illuminants and metamerism — the second exam every match must

pass

Two dyeings with different dye sets can share a reflectance match under D65 daylight yet diverge under TL84 store lighting or illuminant A, because their spectra are only approximately equal. A match prediction is therefore reported with: ΔE under the primary illuminant, ΔE under each secondary illuminant, and a **metamerism index**. The strongest defence is chemical: match with the same dye class as the standard (Topic 1) so the spectra have the same shape, not merely the same three-wavelength sums.

5. Choosing among candidate recipes

CRITERION	WHY IT DECIDES
Cost	Recipes fitting equally well can differ twofold in dye cost; compute cost per kg of fabric for each candidate (Topic 2).
Metamerism	Lowest index across the buyer's illuminants wins ties.
Compatibility	Components must exhaust and fix together; a mathematically perfect recipe from incompatible dyes dyes unlevel and shifts hue on the fibre.
Build-up headroom	Solutions sitting on the flat of a build-up curve cannot be corrected in bulk; prefer recipes in the responsive region.
Fastness	The set must meet the full fastness specification, not just the colour.

6. Glossary — key terms

- **Unit K/S:** a dye's calibrated K/S contribution per unit concentration at a stated wavelength.
- **Least squares:** fitting concentrations to minimise total squared error over all wavelengths.
- **ΔE:** the CIELAB colour difference between trial and standard.
- **Metamerism index:** the colour difference that appears when the illuminant changes.
- **Gamut:** the set of colours a given dye combination can reach.

7. Key points

- K/S additivity converts matching into linear equations; three dyes need at least three wavelengths, production uses the full spectrum.
- Always verify a solved recipe by re-substitution, and a dyed trial by ΔE under every specified illuminant.

- Among equal-fitting recipes, choose by cost, metamerism, compatibility and correctability — not by fit alone.

8. Summary

The worked system in this sheet — unit K/S table, three equations, solution 1.05/0.79/0.46% owf, ΔE 1.07 demanding one correction — is computer colour matching in miniature. The computer adds wavelengths, speed and optimisation; it does not add judgement, which remains the colorist's: the right dye set, the honest calibration, and the refusal to pass a match that only passes under one lamp.

9. Practice

1. A standard has K/S = 1.87 at its λ_{max} (R = 0.18). A single dye's unit K/S there is 1.44 per 1% owf. Predict the concentration, ignoring other dyes' contributions.
2. Standard: L* 60.0, a* −8.0, b* 22.0. Trial: L* 59.2, a* −7.4, b* 21.1. Compute ΔE and state whether it passes a 1.0 tolerance.
3. Two candidate recipes both give ΔE 0.5 under D65. Recipe A costs ৳31/kg fabric with metamerism index 1.8 under TL84; recipe B costs ৳36/kg with index 0.7. Which should be issued, and why?
4. Why does a match-prediction system need the substrate's own K/S (undyed) as well as the dyes' unit values?

Answer key

1. c = 1.87/1.44 = 1.30% owf.
2. ΔL* = 0.8, Δa* = −0.6, Δb* = 0.9 → ΔE = √(0.64 + 0.36 + 0.81) = √1.81 = 1.35. It fails a 1.0 tolerance.
3. Recipe B. The ৳5/kg saving on A is smaller than the cost of one rejected lot under store lighting; the low metamerism index protects the match across the buyer's illuminants.
4. Measured K/S of a dyeing includes the substrate's own absorption and scattering; unit dye values are obtained by subtracting the undyed substrate's contribution, and predictions must add it back — omitting it overstates predicted depth, worst in pale shades.